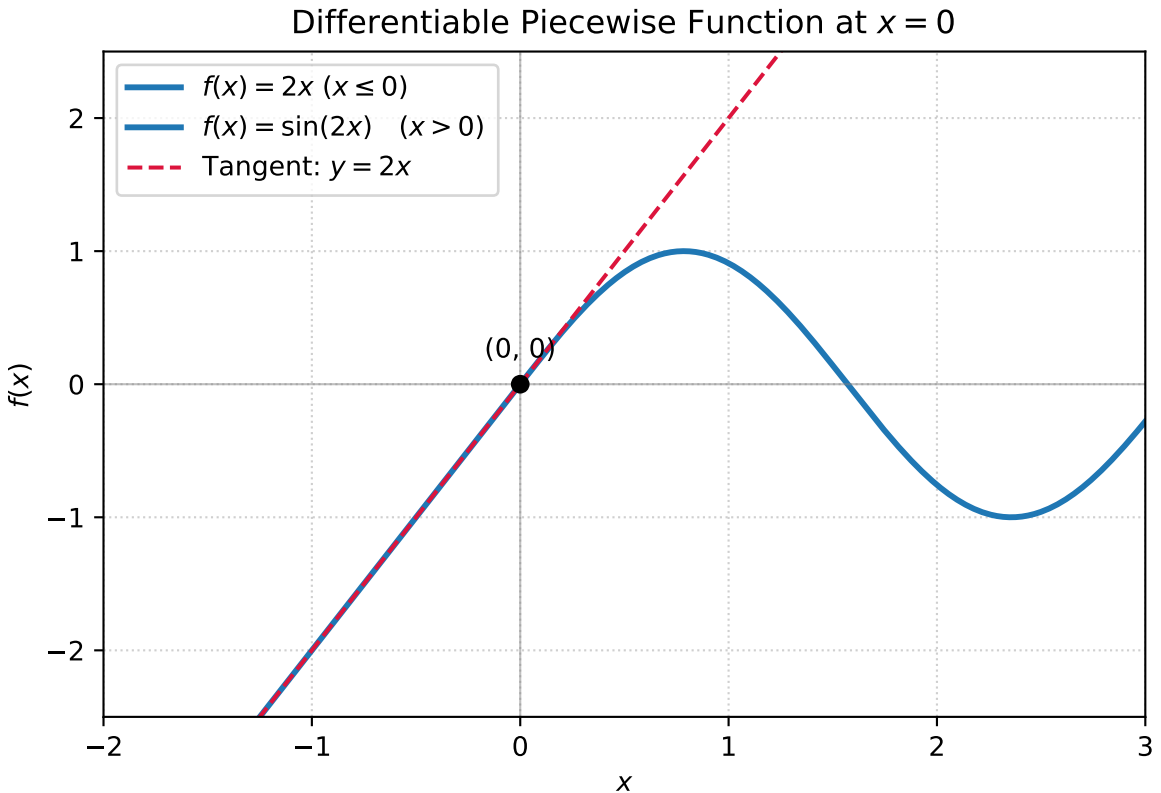
Differentiable Piecewise Function at $x = 0$
$f(x) = 2x$ $(x \leq 0)$
$f(x) = \sin(2x)$ $(x > 0)$
Tangent: $y = 2x$
(0, 0)
$f(x)$
x
2
1
0
−1
−2
−2
−1
0
1
2
3